

Documentar cómo verificaron que systemd se encuentra activo

Systemd como primer proceso

ps -p 1 -o pid,comm,args

```
-bash-5.3# ps -p 1 -o pid,comm,args
  PID COMMAND
    1 systemd
-bash-5.3#
```

ps tree

```
-bash-5.3# pstree
systemd--bash--pstree
        |--dbus-daemon
        |--sshd
        |--systemd-journal
        |--systemd-logind
        |--systemd-network
        |--systemd-nsresou--5*[systemd-nsresou]
        |--systemd-oomd
        |--systemd-resolve
        |--systemd-timesyn--{systemd-timesyn}
        |--systemd-udev
-bash-5.3# _
```

systemctl status

```

• LFSito
  State: running
  Units: 281 loaded (incl. loaded aliases)
  Jobs: 0 queued
  Failed: 0 units
  Since: Thu 2026-02-12 14:36:32 -03; 4min 23s ago
  systemd: 257.8
  Tainted: unmerged-bin
  CGroup: /
    └─init.scope
       └─1 /sbin/init
          └─system.slice
             └─dbus.service
                └─238 /usr/bin/dbus-daemon --system --address=systemd: --nofork
             └─sshd.service
                └─240 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
             └─system-getty.slice
                └─getty@tty1.service
                   └─239 -bash
                      └─282 systemctl status
                         └─283 less
             └─systemd-journald.service
                └─119 /usr/lib/systemd/systemd-journald
             └─systemd-logind.service

```

lines 1-24

Mostrar al menos un servicio levantado/gestionado con systemd

systemctl status sshd

```

-bash-5.3# systemctl status sshd
• sshd.service - OpenSSH Daemon
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena>
  Active: active (running) since Thu 2026-02-12 14:36:43 -03; 5min ago
  Invocation: 32ce32d94a1c43b49f1f1fa795f67a36
  Main PID: 240 (sshd)
  Tasks: 1 (limit: 23610)
  Memory: 2.4M (peak: 2.6M)
  CPU: 142ms
  CGroup: /system.slice/ssh.service
          └─240 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 12 14:36:43 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 12 14:36:44 LFSito sshd[240]: Server listening on 0.0.0.0 port 22.
Feb 12 14:36:44 LFSito sshd[240]: Server listening on :: port 22.
lines 1-14/14 (END)

```

Logs

journalctl -u sshd

```

Jan 24 15:31:28 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 24 15:31:29 LFSito sshd[235]: Server listening on 0.0.0.0 port 22.
Jan 24 15:31:29 LFSito sshd[235]: Server listening on :: port 22.
-- Boot 0da106bf1133462c9e7c07e2308c3b2e --
Jan 24 15:39:15 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 24 15:39:16 LFSito sshd[236]: Server listening on 0.0.0.0 port 22.
Jan 24 15:39:16 LFSito sshd[236]: Server listening on :: port 22.
Jan 24 15:45:56 LFSito sshd-session[275]: Received disconnect from 192.168.0.11>
Jan 24 15:45:56 LFSito sshd-session[275]: Disconnected from user asdy 192.168.0>
-- Boot a79ce60593f8492a82ca5b338345a823 --
Jan 31 10:08:50 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 31 10:08:51 LFSito sshd[235]: Server listening on 0.0.0.0 port 22.
Jan 31 10:08:51 LFSito sshd[235]: Server listening on :: port 22.
-- Boot fadeafed690f442694faa3e12dc9e7e0 --
Feb 04 20:17:44 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 04 20:17:44 LFSito sshd[230]: Server listening on 0.0.0.0 port 22.
Feb 04 20:17:44 LFSito sshd[230]: Server listening on :: port 22.
-- Boot 4c1061a7ce1d48afa23b7f20ab93bd88 --
Feb 06 12:06:30 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 06 12:06:31 LFSito sshd[238]: Server listening on 0.0.0.0 port 22.
Feb 06 12:06:31 LFSito sshd[238]: Server listening on :: port 22.
-- Boot 1559bf12789c49498dee871603ce92bb --
Feb 06 12:15:52 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 06 12:15:52 LFSito sshd[238]: Server listening on 0.0.0.0 port 22.
lines 1-24_

```

Configuración de la unidad

cat /lib/systemd/system/sshd.service

```

-bash-5.3# cat /lib/systemd/system/sshd.service
[Unit]
Description=OpenSSH Daemon

[Service]
ExecStart=/usr/sbin/sshd -D
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=always

[Install]
WantedBy=multi-user.target
-bash-5.3# _

```

Lista de servicios activos

systemctl list-units --type=service --state=running

```

KillMode=process
Restart=always

[Install]
WantedBy=multi-user.target
-bash-5.3# systemctl list-units --type=service --state=running
  UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
  dbus.service                       loaded active running D-Bus System Message Bus
  getty@tty1.service                 loaded active running Getty on tty1
  sshd.service                       loaded active running OpenSSH Daemon
  systemd-journald.service           loaded active running Journal Service
  systemd-logind.service              loaded active running User Login Management
  systemd-networkd.service            loaded active running Network Configuration
  systemd-nsresourced.service         loaded active running Namespace Resource Manager
  systemd-oomd.service                loaded active running Userspace Out-Of-Memory (OOM)
  systemd-resolved.service            loaded active running Network Name Resolution
  systemd-timesyncd.service           loaded active running Network Time Synchronization
  systemd-udev.service                loaded active running Rule-based Manager for Devices and Units

Legend: LOAD    + Reflects whether the unit definition was properly loaded.
          ACTIVE + The high-level unit activation state, i.e. generalization of SUB.
          SUB    + The low-level unit activation state, values depend on unit type.

11 loaded units listed.
lines 1-18/18 (END)

```

Sytemd vs Sysvinit

Al utilizar systemd como sistema init en este proyecto de LFS, se puede contrastar su funcionamiento con lo que ocurriría si se hubiera utilizado Sysvinit tradicional. La diferencia más evidente se observa en el proceso de arranque: con systemd, los servicios se inician de manera paralela, lo que en teoría reduce significativamente el tiempo de boot, mientras que con Sysvinit cada servicio esperaría a que el anterior terminara de iniciar antes de comenzar. Sobre la gestión de servicios, systemd centraliza todo mediante el comando systemctl, permitiendo iniciar, detener o verificar el estado de cualquier servicio con este. En cambio, Sysvinit requeriría ejecutar scripts individuales ubicados en /etc/init.d/ con diferentes implementaciones según cada servicio. El manejo de logs también difiere notablemente. Systemd implementa journalctl para consultar todos los registros del sistema desde este mismo. Las dependencias entre servicios en systemd se declaran explícitamente en los archivos .service mediante directivas como After= o Requires=, mientras que en Sysvinit estas dependencias se manejarían mediante números de secuencia (S01, S02, etc.) que resultan menos claros y más propensos a errores de configuración. Aunque Sysvinit es conceptualmente más simple por basarse en scripts de shell convencionales, systemd ofrece capacidades más robustas para sistemas modernos que requieren mayor control y eficiencia en la gestión de servicios.

Host Capturas

Capturas Sobre información del Host

```
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 4
On-line CPU(s) list: 0-3
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz
BIOS Model name: CPU @ 0.0GHz
BIOS CPU family: 0
CPU family: 6
Model: 94
Thread(s) per core: 1
Core(s) per socket: 4
Socket(s): 1
Stepping: 3
BogoMIPS: 5184.00
Flags: fpu vme de pse tsc mrr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_
tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmuldq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave
aux_16c rdtscd hypervisorlahf_lm abm 3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush_
_lld arch_capabilities

Virtualization features:
Hypervisor vendor: KVM
Virtualization type: full
Caches (sum of all):
L1d: 128 KiB (4 instances)
L1i: 128 KiB (4 instances)
L2: 1 MiB (4 instances)
L3: 24 MiB (4 instances)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-3
Vulnerabilities:
Gather data sampling: Unknown: Dependent on hypervisor status
Indirect target selection: Mitigation: Aligned branch/return thunks
Itlb multihit: Not affected
L1tf: Mitigation: PTE Inversion
Mds: Mitigation: Clear CPU buffers; SMT Host state unknown
Meltdown: Mitigation: PTI
Mmio state data: Mitigation: Clear CPU buffers; SMT Host state unknown
Reg file data sampling: Not affected
Retbleed: Vulnerable
Spec rstack overflow: Not affected
Spec store bypass: Vulnerable
Spectre v1: Mitigation: usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation: Retpolines; STIBP disabled; RSB filling; PBSRB-eIBRS Not affected; BHI Retpoline
Srbds: Unknown: Dependent on hypervisor status
Tsx: Not affected
Tsx async abort: Not affected
[root@localhost ~]#
```

```
[root@localhost ~]# free -h
              total        used        free      shared  buff/cache   available
Mem:           18Gi         555Mi         18Gi          10Mi        198Mi        18Gi
Swap:          11Gi           0B          11Gi

[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb5       20G   3.2G   17G   16% /
devtmpfs        9.5G     0   9.5G    0% /dev
tmpfs           9.5G     0   9.5G    0% /dev/shm
tmpfs           3.8G    11M   3.8G    1% /run
tmpfs           1.0M     0   1.0M    0% /run/credentials/systemd-journald.service
/dev/sdb2       960M   362M   599M   38% /boot
/dev/sdb4       5.0G   130M   4.9G    3% /home
tmpfs           1.0M     0   1.0M    0% /run/credentials/getty@tty1.service
tmpfs           1.9G   4.0K   1.9G    1% /run/user/0
[root@localhost ~]#
```

Sdb6 siendo la partición del LFS

```
[root@localhost ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      1 14.4G  0 disk
└─sda1       8:1      1 14.4G  0 part
sdb          8:16     0 60.4G  0 disk
├─sdb1       8:17     0   1M  0 part
├─sdb2       8:18     0   1G  0 part /boot
├─sdb3       8:19     0  12G  0 part [SWAP]
├─sdb4       8:20     0   5G  0 part /home
├─sdb5       8:21     0  20G  0 part /
└─sdb6       8:22     0 21.4G  0 part
sr0         11:0     1 1024M  0 rom
[root@localhost ~]#
```